

# Saipranay Masadi

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## SUMMARY

Data Scientist with 4+ years of experience building and deploying ML models in E-commerce, Banking, and Mortgage sectors. Skilled in collaborating with cross-functional teams to deliver solutions that boost revenue, enhance efficiency, and align with business goals. Passionate about leveraging data science innovations to drive actionable insights, operational excellence, and impactful decision-making.

## EDUCATION

- **Stanford University, School of Engineering** *Professional Certificate, AI Program* Jan 2025- present
- **Wilmington University, Master's in computer science-Data Science.** *GPA: 3.64* Aug 2022- Dec2023, New Castle, DE
- **Osmania University, Bachelor's in computer science.** *GPA: 3.59* Aug 2014- Dec 2018, Hyd, India

## RELEVANT EXPERIENCE

- **Data Scientist** Aug 2023 – Present  
*Freddie Mac* *McLean, USA*
  - **Achieved a 40% boost in retrieval accuracy by implementing a pilot RAG pipeline** with Pinecone vector DB, utilizing Query Expansion, Focus Mode, and a three-step retrieval process for precise document and sentence-level retrieval.
  - **Enhanced response generation employing Contrastive In-Context Learning (CICL)**, leveraging demonstration examples and contrastive learning for accurate and contextually relevant outputs.
  - **Reduced average contract review time by 30% (from 5 hours to 3.5 hours)** by fine-tuning LayoutLMv3 using Parameter-Efficient Fine-Tuning QLoRA for advanced entity recognition and clause extraction, efficiently processing over 200,000 unstructured contract documents.
  - Built a scalable ML development pipeline using PostgreSQL for metadata storage, DVC for dataset and artifact versioning, and S3 for storing PDFs and model artifacts.
  - Deployed production-grade ML pipelines on EKS Kubernetes, integrating Traefik for load balancing, MLflow for model tracking, and Fast API for serving models through a web app interface
- **Data Scientist** Jan 2020 – Jul 2022  
*Hexaware Technologies(e-commerce)* *Hyderabad, India*
  - **Increased revenue by 16% through targeted marketing campaigns**, cross-sell and upsell Techniques using customer segmentation.
  - **Prevented \$43K in fraud within six months by implementing a classification model** for magazine subscription fraud detection.
  - **Improved customer service efficiency and saved 100s man hours per month** by developing a BERT-based email classification system.
  - Designed an A/B test to compare standard vs. increased email frequency for the marketing team, leading to data-driven insights for customer engagement.
  - Developed and refined various Deep learning algorithms, including LSTM, BERT, address unsupervised learning objectives.
  - Developed end-to-end CI/CD processes for ML models using Terraform and Ansible, ensuring scalable and reliable deployment pipelines, and monitoring model performance with AWS services and tools like Sage Maker, Lambda, and CloudWatch.
- **Data & Business Analyst** Aug 2018 – Dec 2019  
*Streebo Inc(banking)* *Hyderabad, India*
  - **Led the initial assessment for BASEL 3 to BASEL 4 transition**, guiding the process with expertise in operational risk frameworks. Streamlined BASEL 4 compliance by authorizing BRD and crafting Functional Requirement Documents.
  - **Devised strategies that reduced development efforts by 50%, saving millions** and enhancing financial stability and risk management. Engineered a shift towards an intuitive NCB alphanumeric system, **reducing report generation jobs by 40%.**

## PERSONAL PROJECTS

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- DeepSeek RAGenius: A Localized RAG System for Intelligent PDF Q&A Using DeepSeek and Ollama.
- Efficient GRPO Reasoning Model: Trained a reasoning model using optimized GRPO, reducing VRAM usage by 80% for efficient fine-tuning.
- Bald Classification HairStageNet: Built a CNN-based deep learning model for baldness classification using PyTorch.
- The Augmented LLM: Built an Agent to summaries/refine/make an article and then validate the Data references.
- Artique-Frames Chatbot: Developed an NLP-driven chatbot for enhancing user interaction in e-commerce.
- Topic Generation Tool: Created a content recommendation tool leveraging the Llama 2-7B model.

## TECHNICAL SKILLS

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**Tools/Technologies:** Python, Java, C++, R, SQL, Postgres S Q L , MongoDB, NLP (NLTK, Spacy, Scikit-Learn, BERT, LayoutLM, LLMs, Langchain, Pinecone, Open AI API),Generative AI, Retrieval Augment generation, Prompt Engineering, Deep Learning, Fine tuning, TensorFlow, Keras, PyTorch, Kafka, Rabbit MQ, Spark, Hadoop, Hive, AWS ML, Sage Maker, S3, EC2, ECR, CI/CD, A/B Testing, FAISS, Flask, streamlit, Docker, Kubernetes , Prefect, Airflow, Statistics, Tableau, Hugging Face, Git, GitHub, Jira.